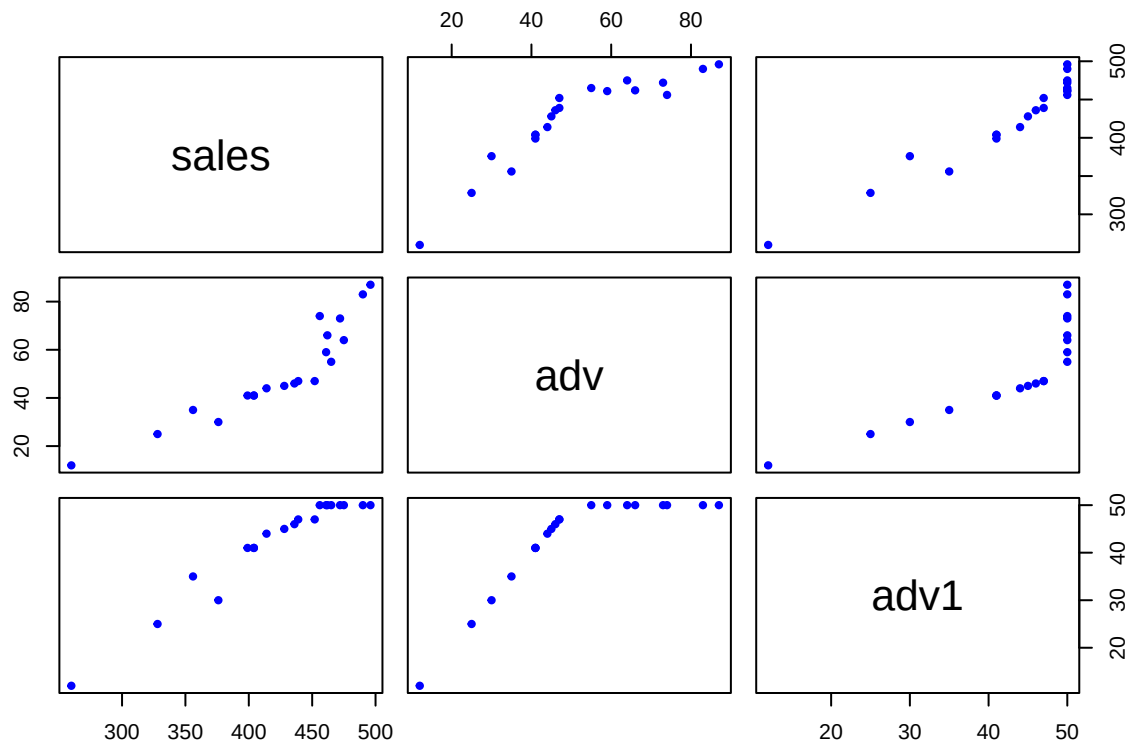


Ad Analysis

Gustavo Cepparo and Milica Cudina

First, we download the data set and take a look at it.

```
data<-read.csv("salesadv.csv",header=TRUE)
attach(data)
#View(data)
plot(data[,4],
      col="blue", pch=20)
```



Just for laughs, let's fit a multiple linear regression.

```
lm.fit.m=lm(sales ~ adv + adv1, data=data)
summary(lm.fit.m)
```

```
##
## Call:
## lm(formula = sales ~ adv + adv1, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.2078  -4.0006   0.1739   6.1054  23.9011
##
## Coefficients:
```

```
##           Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 201.4454    11.6992  17.219 0.00000000000034 ***
## adv         0.9658     0.2398   4.027   0.000875 ***
## adv1        4.0560     0.4571   8.874 0.0000000865618 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.05 on 17 degrees of freedom
## Multiple R-squared:  0.9684, Adjusted R-squared:  0.9647
## F-statistic: 260.6 on 2 and 17 DF,  p-value: 0.0000000000001763

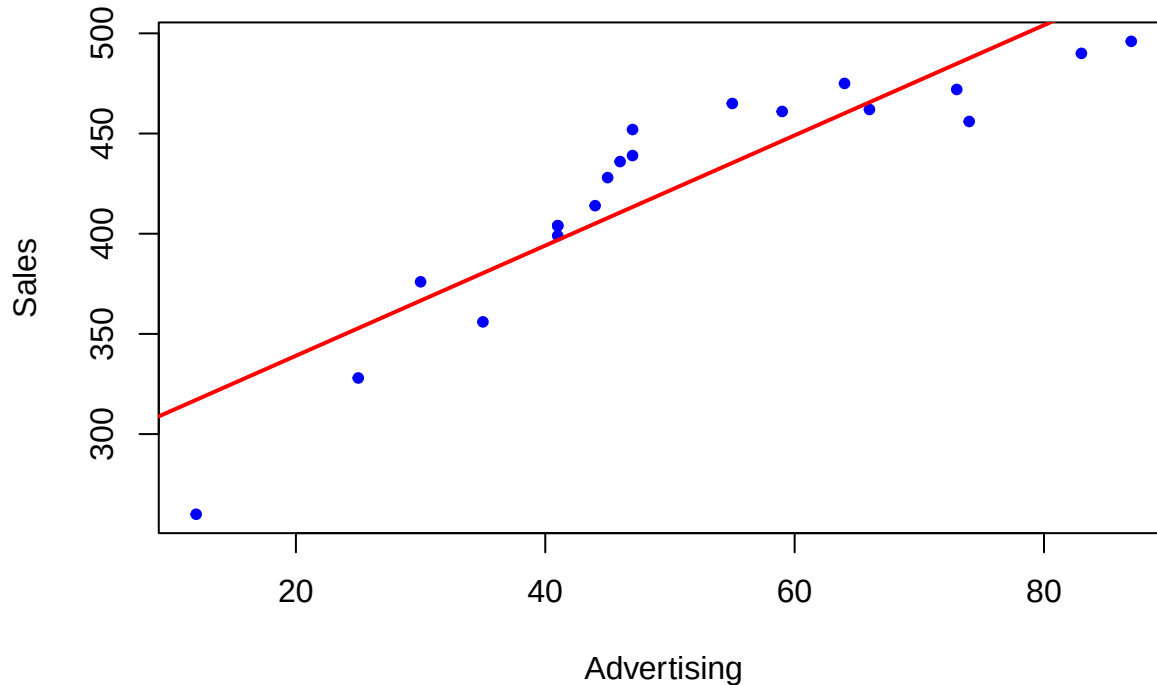
lm.fit.mi=lm(sales ~ adv*adv1, data=data)
summary(lm.fit.mi)
```

```
##
## Call:
## lm(formula = sales ~ adv * adv1, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.9117  -4.5606  -0.2768   6.1266  25.2126
##
## Coefficients:
##           Estimate Std. Error t value      Pr(>|t|)
## (Intercept) 209.115091  25.982946   8.048 0.000000513 ***
## adv         0.481233   1.476049   0.326   0.749
## adv1        3.981928   0.519587   7.664 0.000000964 ***
## adv:adv1     0.008642   0.025956   0.333   0.744
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.35 on 16 degrees of freedom
## Multiple R-squared:  0.9686, Adjusted R-squared:  0.9627
## F-statistic: 164.7 on 3 and 16 DF,  p-value: 0.0000000000003089
```

We are going to focus on just the fit with adv as a single predictor. So, let's look at the simple linear regression.

```
plot(adv, sales,
     pch=20, col="blue",
     main="Dependence of sales on advertising",
     xlab="Advertising",
     ylab="Sales")
lm.fit<-lm(sales ~ adv)
abline(lm.fit, col="red", lwd=2)
```

Dependence of sales on advertising



```
summary(lm.fit)
```

```
##
## Call:
## lm(formula = sales ~ adv)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -57.091 -22.836   7.162  16.226  38.662
##
## Coefficients:
##              Estimate Std. Error t value      Pr(>|t|)
## (Intercept)  284.0915    16.3292    17.40 0.00000000000105 ***
## adv           2.7499     0.3015     9.12 0.00000003615574 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 25.48 on 18 degrees of freedom
## Multiple R-squared:  0.8221, Adjusted R-squared:  0.8122
## F-statistic: 83.17 on 1 and 18 DF,  p-value: 0.00000003616
```

Since there appears to be a difference in the “signal” in the vicinity of 50 in adv, it’s time to consider linear splines. First, we import the library `splines`.

```
library(splines)
```

Then, we do create a fit with `degree=1` - meaning that it’s a linear spline. Note, that we specify a single knot at 50.

```
lm.fit.ls<-lm(sales~bs(adv,knots=c(50),degree=1),data=data)
summary(lm.fit.ls)
```

```
##
## Call:
## lm(formula = sales ~ bs(adv, knots = c(50), degree = 1), data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -21.2078  -4.0006   0.1739   6.1054  23.9011
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                   261.71      8.41    31.12
## bs(adv, knots = c(50), degree = 1)1  190.83      10.92    17.47
## bs(adv, knots = c(50), degree = 1)2  226.56      10.07    22.50
##                                Pr(>|t|)
## (Intercept)                   < 2e-16 ***
## bs(adv, knots = c(50), degree = 1)1 0.0000000000026980 ***
## bs(adv, knots = c(50), degree = 1)2 0.000000000000433 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.05 on 17 degrees of freedom
## Multiple R-squared:  0.9684, Adjusted R-squared:  0.9647
## F-statistic: 260.6 on 2 and 17 DF,  p-value: 0.000000000001763
```

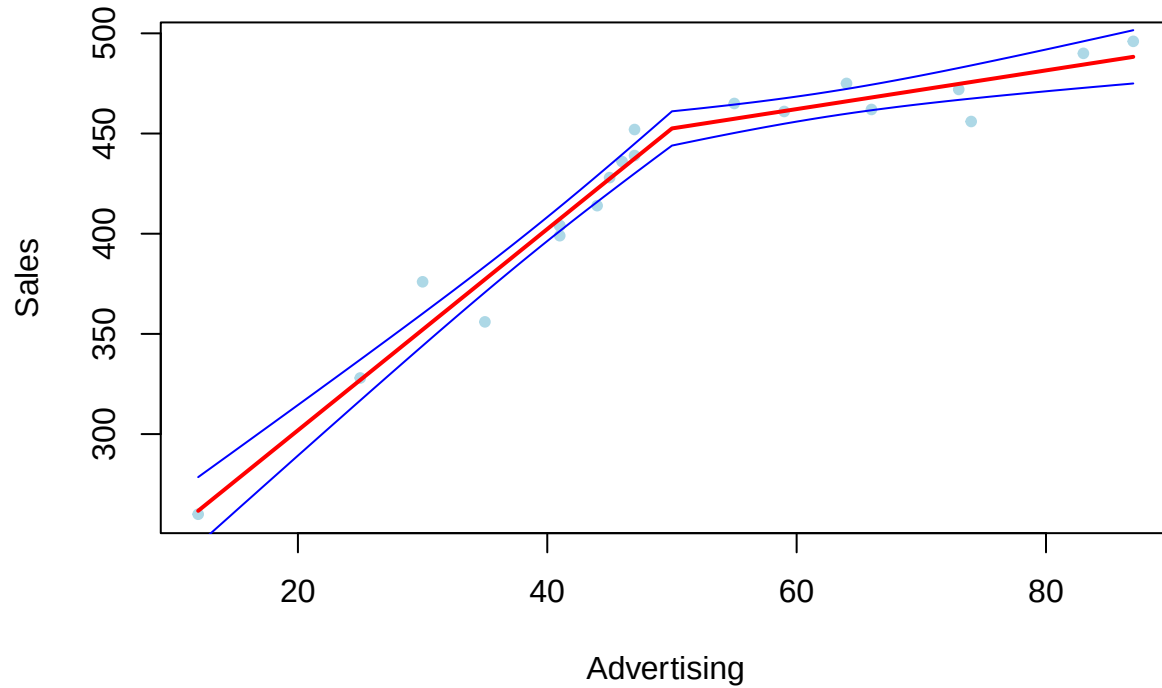
The R^2 is better. What about the visuals?

```
adv.mesh=seq(from=min(adv),to=max(adv), by=0.5)
predictions=predict(lm.fit.ls,newdata=list(adv=adv.mesh),se=T)

plot(adv, sales,
     pch=20, col="lightblue",
     main="Dependence of sales on advertising",
     xlab="Advertising",
     ylab="Sales")

lines(adv.mesh,predictions$fit,col="red", lwd=2)
lines(adv.mesh,predictions$fit+2*predictions$se, col="blue")
lines(adv.mesh,predictions$fit-2*predictions$se, col="blue")
```

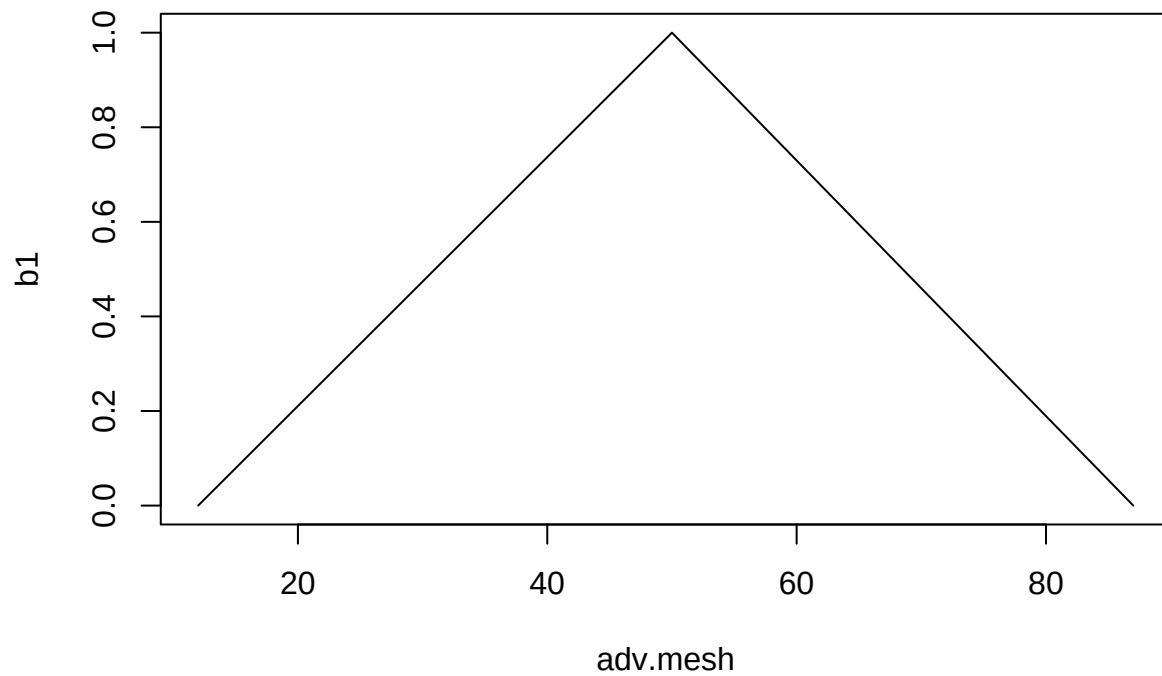
Dependence of sales on advertising



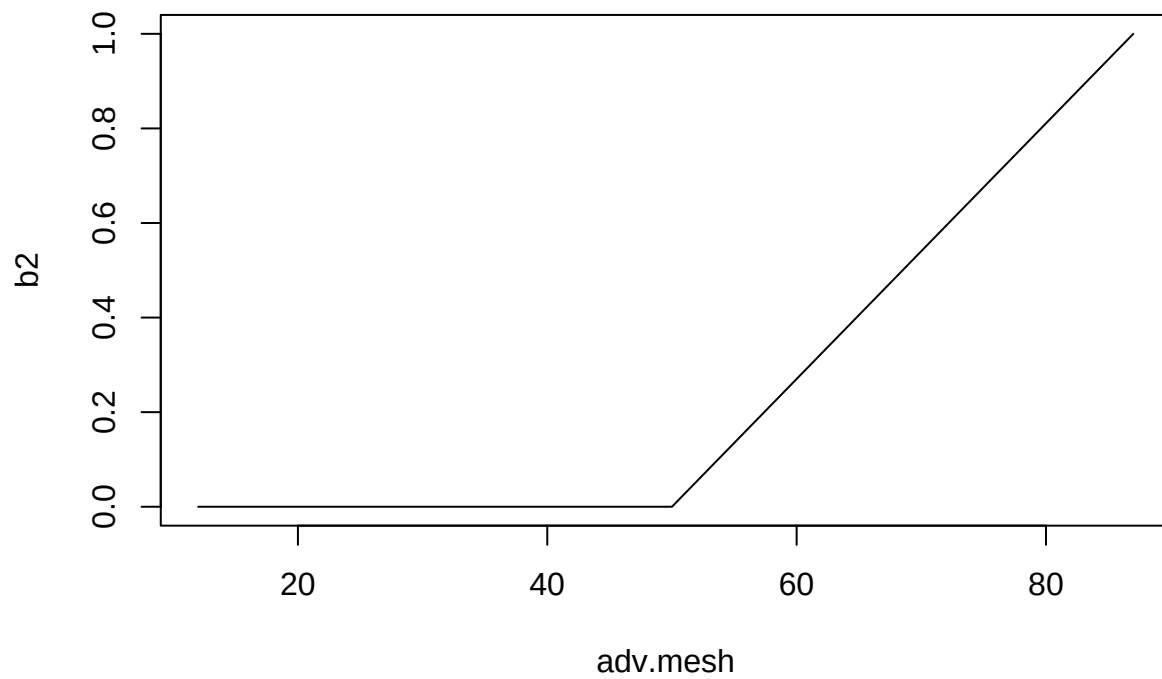
Of course, we might be interested in the basis functions used in this implementation.

```
## returns a basis matrix
b <- bs(adv.mesh, degree = 1, knots = c(50))
#b

#the functions in the basis
b1 <- b[, 1]
b2 <- b[, 2]
plot(adv.mesh, b1, type = "l")
```



```
plot(adv.mesh, b2, type = "l")
```



```
#the coefficients from the summary of `lm`
(betas=summary(lm.fit.ls)$coeff[,1])
```

```
##                (Intercept) bs(adv, knots = c(50), degree = 1)1
##                261.7068                190.8278
## bs(adv, knots = c(50), degree = 1)2
##                226.5612
```

```
plot(adv, sales,
     pch=20, col="lightblue",
```

```
main="Dependence of sales on advertising",  
xlab="Advertising",  
ylab="Sales")
```

```
lines(adv.mesh,betas[1]+betas[2]*b1+betas[3]*b2,col="red", lwd=2)
```

